Lecture 7:
Quantum phase estimation

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Encoding information in phases

In several quantum algorithms information is encoded in the relative phases of a quantum state.

The effect of Hadamard (once again)

$$H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) = \frac{1}{\sqrt{2}}\sum_{y\in 2}(-1)^{xy}|y\rangle$$

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}}\sum_{y\in 2^n}(-1)^{x\cdot y}|y\rangle$$

is to encode information about the value of x into the phases $(-1)^{x\cdot y}$ of basis states $|y\rangle$.

Encoding information in phases

Of course, as a reversible gate, the Hadamard gate also decodes information from phases:

$$
\begin{aligned}
H^{\otimes n}\frac{1}{\sqrt{2^n}}\sum_{y\in 2^n}(-1)^{x\cdot y}|y\rangle &= H^{\otimes n}(H^{\otimes n}|x\rangle) \\
&= (H^{\otimes n}H^{\otimes n})|x\rangle \\
&= I|x\rangle \\
&= |x\rangle
\end{aligned}
$$

Encoding information in phases

In general, phases are complex numbers

$$e^{2\pi i w}$$

for any real $w \in [0, 1[$.

Of course, $H^{\otimes n}$ cannot encode/decode information over such generic phases. The general situation can be described as follows:

The phase estimation problem

Determine a good estimation of the phase parameter w given a general quantum state

$$\frac{1}{\sqrt{2^n}} \sum_{y \in 2^n} e^{2\pi i w y} |y\rangle$$

An algorithm for phase estimation

Notation

$$w \;=\; 0\,.\,x_1x_2\cdots$$

is written in base 2 (i.e. $w = x_1 2^{-1} + x_2 2^{-2} + \cdots$); thus

$$2^k w \;=\; x_1x_2\cdots x_k\,.\,x_{k+1}x_{k+2}\cdots$$

and

$$\begin{aligned} e^{2\pi i(2^k w)} &= e^{2\pi i(x_1x_2\cdots x_k\,.\,x_{k+1}x_{k+2}\cdots)} \\ &= e^{2\pi i(x_1x_2\cdots x_k)}\; e^{2\pi i(0\,.\,x_{k+1}x_{k+2}\cdots)} \\ &= e^{2\pi i(0\,.\,x_{k+1}x_{k+2}\cdots)} \end{aligned}$$

because $e^{2\pi iz} = 1$ for any integer z.

Case A: 1-qubit state and $w = 0.x_1$

$$\begin{aligned}
\frac{1}{\sqrt{2}} \sum_{y \in 2} e^{2\pi i (0.x_1) y} |y\rangle &= \frac{1}{\sqrt{2}} \sum_{y \in 2} e^{2\pi i (\frac{x_1}{2}) y} |y\rangle \\
&= \frac{1}{\sqrt{2}} \sum_{y \in 2} e^{\pi i (x_1 y)} |y\rangle \\
&= \frac{1}{\sqrt{2}} \sum_{y \in 2} (-1)^{x_1 y} |y\rangle \\
&= \frac{1}{\sqrt{2}} (|0\rangle + (-1)^{x_1} |1\rangle)
\end{aligned}$$

Clearly H will decode and retrieve x_1 because

$$H \left(\frac{1}{\sqrt{2}} (|0\rangle + (-1)^{x_1} |1\rangle) \right) = |x_1\rangle$$

Case B: 2-qubit state and $w = 0.x_1x_2$

Observe that

$$\frac{1}{\sqrt{2^2}}\sum_{y\in 2^2} e^{2\pi i(0.x_1x_2)y}|y\rangle = \left(\frac{|0\rangle + e^{2\pi i(0.x_2)}|1\rangle}{\sqrt{2}}\right)\otimes\left(\frac{|0\rangle + e^{2\pi i(0.x_1x_2)}|1\rangle}{\sqrt{2}}\right)$$

which means that x_2, but not x_1, can be retrieved from the first qubit through an application of H.

The phase rotator

$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{4}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i(0.01)} \end{bmatrix}$$

where 0.01 is in base 2 (thus, equal to 2^{-2}).

Case B: 2-qubit state and $w = 0.x_1x_2$

Taking $x_2 = 1$ and applying the inverse of the phase rotator to the second qubit, yields

$$R_2^{-1}\left(\frac{|0\rangle + e^{2\pi i(0.x_1 1)}|1\rangle}{\sqrt{2}}\right) = \begin{bmatrix} 1 & 0 \\ 0 & e^{-2\pi i(0.01)} \end{bmatrix}\left(\frac{|0\rangle + e^{2\pi i(0.x_1 1)}|1\rangle}{\sqrt{2}}\right)$$

$$= \frac{|0\rangle + e^{2\pi i(0.x_1 1 - 0.01)}|1\rangle}{\sqrt{2}}$$

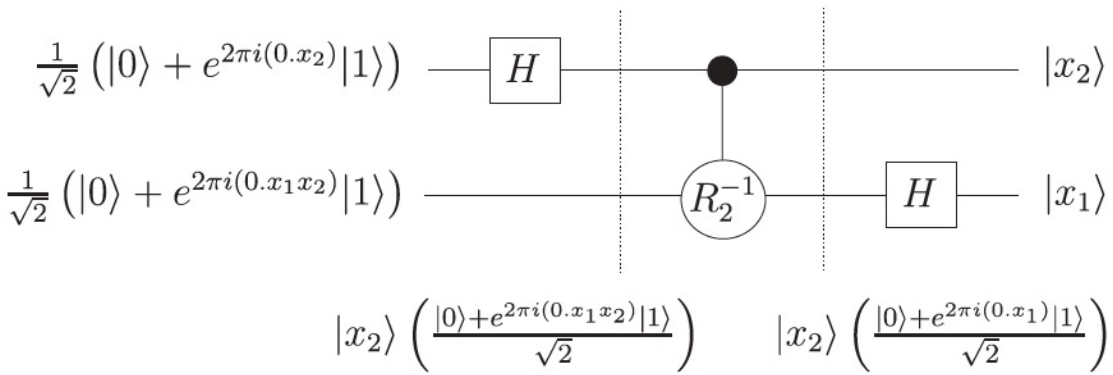
$$= \frac{|0\rangle + e^{2\pi i(0.x_1)}|1\rangle}{\sqrt{2}}$$

Concluding

- x_1 can now be determined by an application of H, as before.
- Moreover, the decision to apply R_2^{-1} before the application of H depends on x_2 being 1 or 0, respectively.
- Thus, to find $w = 0.x_1x_2$ it is enough to apply a controlled version of R, precisely controlled by the state of the first qubit.

Case B: 2-qubit state and $w = 0.x_1x_2$

The circuit



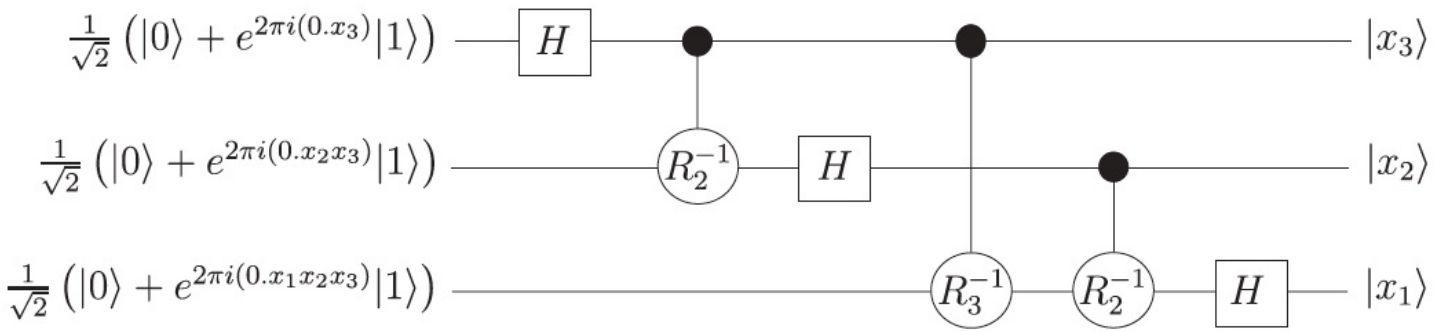
Case C: 3-qubit state and $w = 0.x_1x_2x_3$

The state is now

$$\frac{1}{\sqrt{2^3}} \sum_{y \in 2^3} e^{2\pi i (0.x_1x_2x_3)y} |y\rangle \; =$$

$$= \left(\frac{|0\rangle + e^{2\pi i (0.x_3)} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{2\pi i (0.x_2x_3)} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{2\pi i (0.x_1x_2x_3)} |1\rangle}{\sqrt{2}} \right)$$

In this case the third qubit has to conditionally rotate both x_2 and x_3, leading to the following circuit



Going generic

Gate R_3 in the circuit is an instance of a 1-qubit phase rotator

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & \omega_k = e^{\frac{2\pi i}{2^k}} \end{bmatrix}$$

whose inverse acts as

$$\begin{aligned} R_k^{-1}|0\rangle &= |0\rangle \\ R_k^{-1}|1\rangle &= e^{-2\pi i(0.0\cdots 1)}|1\rangle \end{aligned}$$

with 1 in $0.0\cdots 1$ appearing in position k.

Going generic

The output state of the circuit is

$$|x_3 x_2 x_1\rangle$$

Thus, relabelling the qubits in reverse order, this provides an efficient circuit to estimate the phase (actually, to give a totally accurate estimation ...), by computing

$$\frac{1}{\sqrt{2^n}} \sum_{y \in 2^n} e^{2\pi i (\frac{x}{2^n}) y} |y\rangle \quad \rightsquigarrow \quad |x\rangle$$

From $\sum$ to $\bigotimes$

Let $|x\rangle = |x_1 \cdots x_n\rangle$. A basic observation used before (but not proved yet) claims that

$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i (\frac{x}{2^n})y} |y\rangle$$

is equal to

$$\left(\frac{|0\rangle + e^{2\pi i (0.x_n)}|1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + e^{2\pi i (0.x_{n-1}x_n)}|1\rangle}{\sqrt{2}}\right) \cdots \otimes \cdots \left(\frac{|0\rangle + e^{2\pi i (0.x_1 x_2 \cdots x_n)}|1\rangle}{\sqrt{2}}\right)$$

$$|x\rangle = |x_1 x_2\rangle$$

$$\begin{aligned}
\frac{1}{2}\sum_{y=0}^{3} e^{2\pi i(\frac{x}{2^2})y}|y\rangle &= \frac{1}{2}\sum_{y=0}^{3} e^{2\pi i x y 2^{-2}}|y\rangle \\
&= \frac{1}{2}\sum_{y_1,y_2=0}^{1} e^{2\pi i x(y_1 2^{-1}+y_2 2^{-2})}\,|y_1 y_2\rangle \\
&= \cdots
\end{aligned}$$

The last step holds because, in general,

$$\frac{y}{2^n} = \sum_{j=1}^{n} y_j 2^{-j}$$

$$|x\rangle = |x_1 x_2\rangle$$

$$\begin{aligned}
\cdots &= \frac{1}{2} \sum_{y_1, y_2 = 0}^{1} (e^{2\pi i x y_1 2^{-1}} |y_1\rangle \otimes e^{2\pi i x y_2 2^{-2}} |y_2\rangle) \\
&= \frac{1}{2} \sum_{y_1 = 0}^{1} (e^{2\pi i x y_1 2^{-1}} |y_1\rangle \otimes \sum_{y_2 = 0}^{1} e^{2\pi i x y_2 2^{-2}} |y_2\rangle) \\
&= \frac{(|0\rangle + e^{2\pi i x 2^{-1}} |1\rangle)}{\sqrt{2}} \otimes \frac{(|0\rangle + e^{2\pi i x 2^{-2}} |1\rangle)}{\sqrt{2}} \\
&= \frac{(|0\rangle + e^{2\pi i (x_1 . x_2)} |1\rangle)}{\sqrt{2}} \otimes \frac{(|0\rangle + e^{2\pi i (0. x_1 x_2)} |1\rangle)}{\sqrt{2}} \\
&= \frac{(|0\rangle + e^{2\pi i (0. x_2)} |1\rangle)}{\sqrt{2}} \otimes \frac{(|0\rangle + e^{2\pi i (0. x_1 x_2)} |1\rangle)}{\sqrt{2}}
\end{aligned}$$

because, $e^{2\pi i (a.b)} = e^{2\pi i a} e^{2\pi i (0.b)} = e^{2\pi i (0.b)}$

Are we done?

The phase estimation algorithm works only when the phase is of the form $w = 0.x_1x_2\cdots x_n$, i.e. $\frac{x}{2^n}$ for some integer x

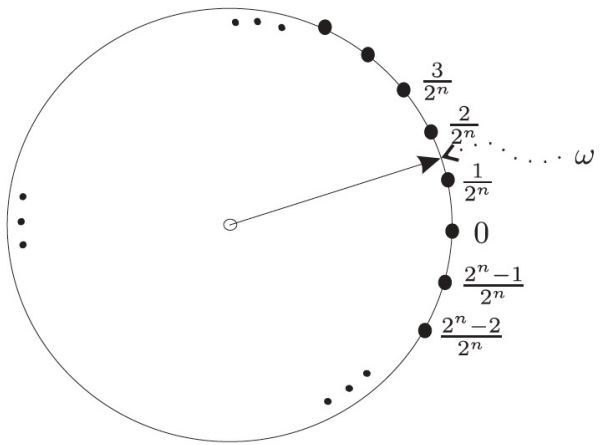
However, it can be shown that, for an arbitrary w, the algorithm will compute x such that $\frac{x}{2^n}$ is closest to w with high probability.

The question

What is the error emerging when w is not an integer multiple of $\frac{1}{2^n}$?

Are we done?

Consider w an integer not multiple of $\frac{1}{2^n}$, and let $\hat{w}$ be the nearest integer multiple of $\frac{1}{2^n}$ to w, i.e. $\hat{w} = \frac{\hat{x}}{2^n}$ is the closest number of this form to w.



Theorem

The phase estimation algorithm returns $\hat{x}$ with probability at least $\frac{4}{\pi^2}$, i.e. the algorithm outputs an estimate $\hat{x}$ with the given probability such that

$$\left| \frac{\hat{x}}{2^n} - w \right| \leq \frac{1}{2^{n+1}}$$

Are we done?

Theorem

$$\text{If} \quad \frac{x}{2^n} \leq w \leq \frac{x+1}{2^n}$$

The phase estimation algorithm returns either x or $x+1$ with probability at least $\frac{8}{\pi^2}$ i.e. the algorithm outputs an estimate $\hat{x}$ with the given probability such that

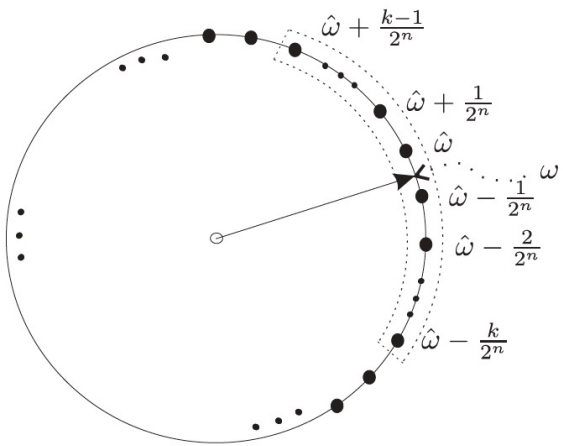
$$\left| \frac{\hat{x}}{2^n} - w \right| = \frac{1}{2^n}$$

The reverse question

How many qubits are required to get w accurate to n bits, with a probability p below a certain level?

Actually, the crucial choice is the value of n (number of qubits used) to ensure the estimation is close enough.

For $p = 1 - \frac{1}{2(k-1)}$, the algorithm returns one of the 2k closest integer multiples of $\frac{1}{2^n}$, i.e.



which means that $|w - \hat{w}| \leq \frac{k}{2^n}$.

The reverse question

Thus, to estimate $\hat{w}$ such that $|w - \hat{w}| \leq \frac{1}{2^r}$ with probability at least

$$1 - \frac{1}{2^m}$$

the maximum number of qubits required is

$$n = r + m + 1$$

- In practice a much smaller error is obtained: for example, with probability at least $\frac{8}{\pi^2}$, the error will be at most

$$\frac{1}{2^{r+m}}$$